

SIMATS ENGINEERING

CO3- AT2 - Design Task - Medium

COURSE NAME: Advance Wireless Communication Systems /

5G / 6G Communication Systems

COURSE CODE: ECA5607

COURSE OUTCOME COVERED

CO: 3 - *Design spectrum access strategies, waveform generation methods, and multiple access techniques for efficient wireless transmission systems. (BL6)*

Assessment Tool Weightage: 35%

Short description about assessment: Students design a system / model based on given requirements to assess creativity and application skills.

Dr. Tamilselvi M

Code of Conduct:

I S.SRIKRISHNAN 192412008 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

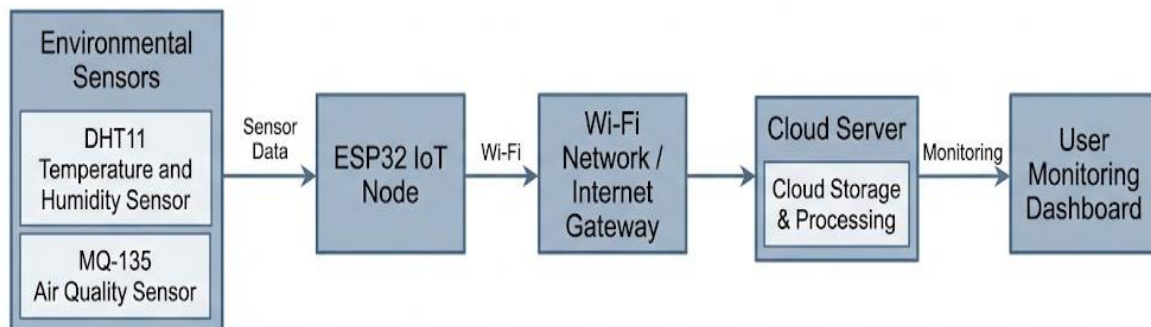
Signature of the student:

RUBRICS FOR CALCULATION:

Design Task					
Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Design	20%	Demonstrates deep understanding of design objectives, constraints, and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Lacks understanding of the design context
Evaluation of Design	25%	Provides in-depth critique identifying strengths, weaknesses, and design gaps	Identifies key issues with reasonable analysis	Limited critique; mostly descriptive feedback	No meaningful critique provided
Justification	20%	Supports critique with strong reasoning, evidence, and relevant principles	Provides justification with some supporting evidence	Weak or unclear justification	No justification for feedback
Suggestions for Improvement	20%	Offers innovative, practical, and well-justified improvement suggestions	Provides useful suggestions with minor gaps	Suggestions are limited or partially relevant	No constructive suggestions provided
Communication	15%	Communicates feedback clearly, professionally, and engages actively in discussion	Communicates well with minor issues	Communication lacks clarity or engagement	Poor communication and minimal participation

31. Design an IoT communication flow for periodic environmental sensing. (BL6)

Design and Implementation of an IoT Communication Flow for Periodic Environmental Sensing



Parameters Used

The following parameters are used in the MATLAB simulation of **IoT Periodic Environmental Sensing**:

- **Number of Sensor Readings (N) = 20:** The system collects 20 environmental readings during the simulation.
- **Sampling Interval (T) = 10 seconds:** Sensors take a new reading every 10 seconds.
- **Temperature:** Simulated around an average of **25°C** with small random variations.
- **Humidity:** Simulated around an average of **60%** with random variations.
- **Air Quality:** Simulated around an average value of **300** with random variations.
- **Temperature Range:** Limited between **15°C and 40°C** to maintain realistic values.
- **Humidity Range:** Limited between **30% and 90%**.
- **Air Quality Minimum:** Restricted to a minimum value of **100**.
- **Random Noise:** `randn()` is used to introduce natural variations in the sensor readings.

These parameters help simulate **periodic environmental data collection and monitoring** in an IoT system.

MATLAB CODE:

```
clc;

clear;

close all;

% IoT Periodic Environmental Sensing Simulation

N = 20;

T = 10;

time = (0:N-1) * T;

temperature = 25 + 2 * randn(1,N);

humidity = 60 + 5 * randn(1,N);

airQuality = 300 + 30 * randn(1,N);

temperature = max(min(temperature,40),15);

humidity = max(min(humidity,90),30);

airQuality = max(airQuality,100);

fprintf('IoT Environmental Monitoring Simulation\n');

fprintf('-----\n');

fprintf('Time(s)\tTemperature(C)\tHumidity(%%)\tAir Quality\n');

for i = 1:N

    fprintf('%d\t%.2f\t%.2f\t%.2f\n', ...

        time(i), temperature(i), humidity(i), airQuality(i));

end

figure;

plot(time, temperature, '-o');

grid on;

xlabel('Time (seconds)');

ylabel('Temperature (C)');

title('Periodic Temperature Monitoring');

figure;

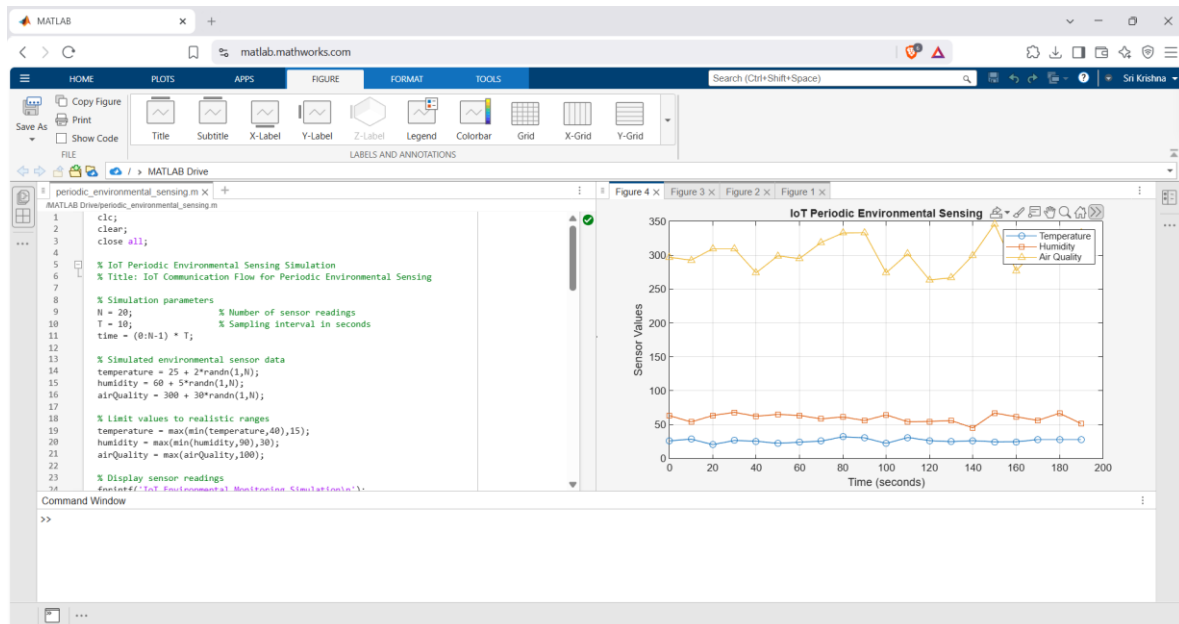
plot(time, humidity, '-o');

grid on;
```

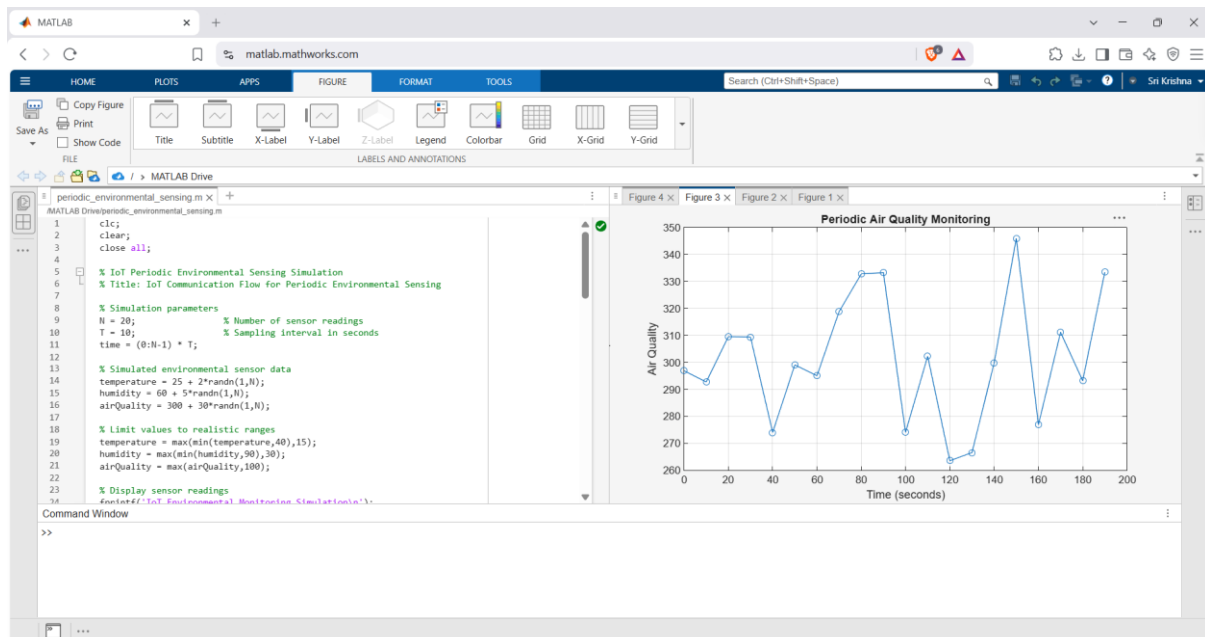
```
xlabel('Time (seconds)');  
ylabel('Humidity (%)');  
title('Periodic Humidity Monitoring');  
figure;  
plot(time, airQuality, '-o');  
grid on;  
xlabel('Time (seconds)');  
ylabel('Air Quality');  
title('Periodic Air Quality Monitoring');  
figure;  
plot(time, temperature, '-o');  
hold on;  
plot(time, humidity, '-s');  
plot(time, airQuality, '-^');  
grid on;  
xlabel('Time (seconds)');  
ylabel('Sensor Values');  
title('IoT Periodic Environmental Sensing');  
legend('Temperature','Humidity','Air Quality');
```

SIMULATION OUTPUT:

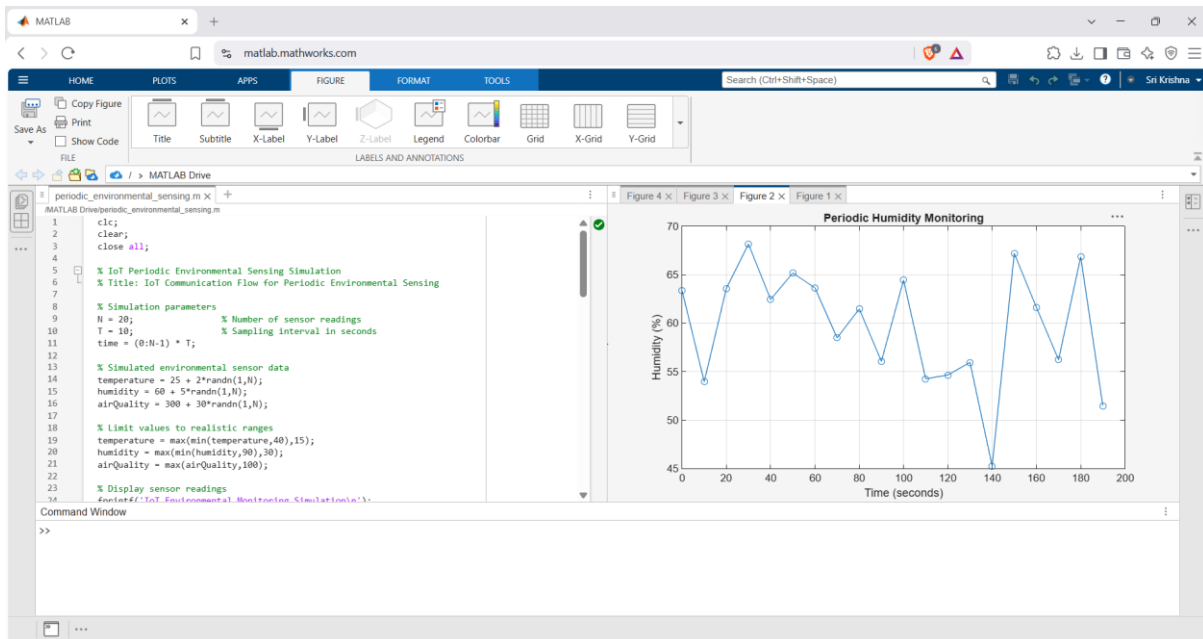
i. IoT Periodic Environmental Sensing



ii. Periodic Air Quality Monitoring



iii. Periodic Humidity Monitoring



iv. Periodic Temperature Monitoring

